

Navigating cyber risks with Microsoft Security Exposure Management



Charting a course with Microsoft Security Exposure Management

In today's interconnected digital ecosystem, organizations navigate a sea of evolving cyberthreats. Microsoft Security Exposure Management serves as your compass, map, and radar system combined—providing continuous visibility and proactive defense across hybrid environments.

Foundational capabilities of Microsoft Security Exposure Management

Your navigational toolkit consists of three core capabilities that anchor every security journey:

Attack surface management

Continuously discovers more assets through unified visibility across:

- Hybrid infrastructure (on-premises/cloud)
- Identity ecosystems (Microsoft Entra ID, Microsoft 365)
- Third-party SaaS applications

Attack path analysis

Visualizes critical attack vectors more effectively than traditional scanners using:

- Hybrid path modeling (on-premises to cloud)
- Discretionary access control list permission chain analysis
- Choke point identification

Unified exposure insights

Proactively drive, manage, and govern security posture programs attached to risks like:

- Business email compromise (BEC) defense
- Ransomware resilience
- Cloud posture optimization

Supporting your proactive cyber-defense journey

Modern cyberthreats require constant vigilance. Microsoft Security Exposure Management provides a comprehensive approach that balances both reactive and proactive security measures. By identifying immediate risks while building sustainable security programs, organizations can stay ahead of evolving threats and strengthen their security posture over time.

Microsoft Security Exposure Management enables organizations to:



Map the attack surface

- Continuous attack surface monitoring
- Predictive exposure insights
- Threat-led security initiatives



Assess connected risks through the asset relationships

- Automated attack path simulations
- Blast radius analysis
- Purple team exercise integration



Mobilize and remediate risks effectively

- Choke point remediation playbooks
- Cross-team exposure dashboards
- Compliance progress tracking



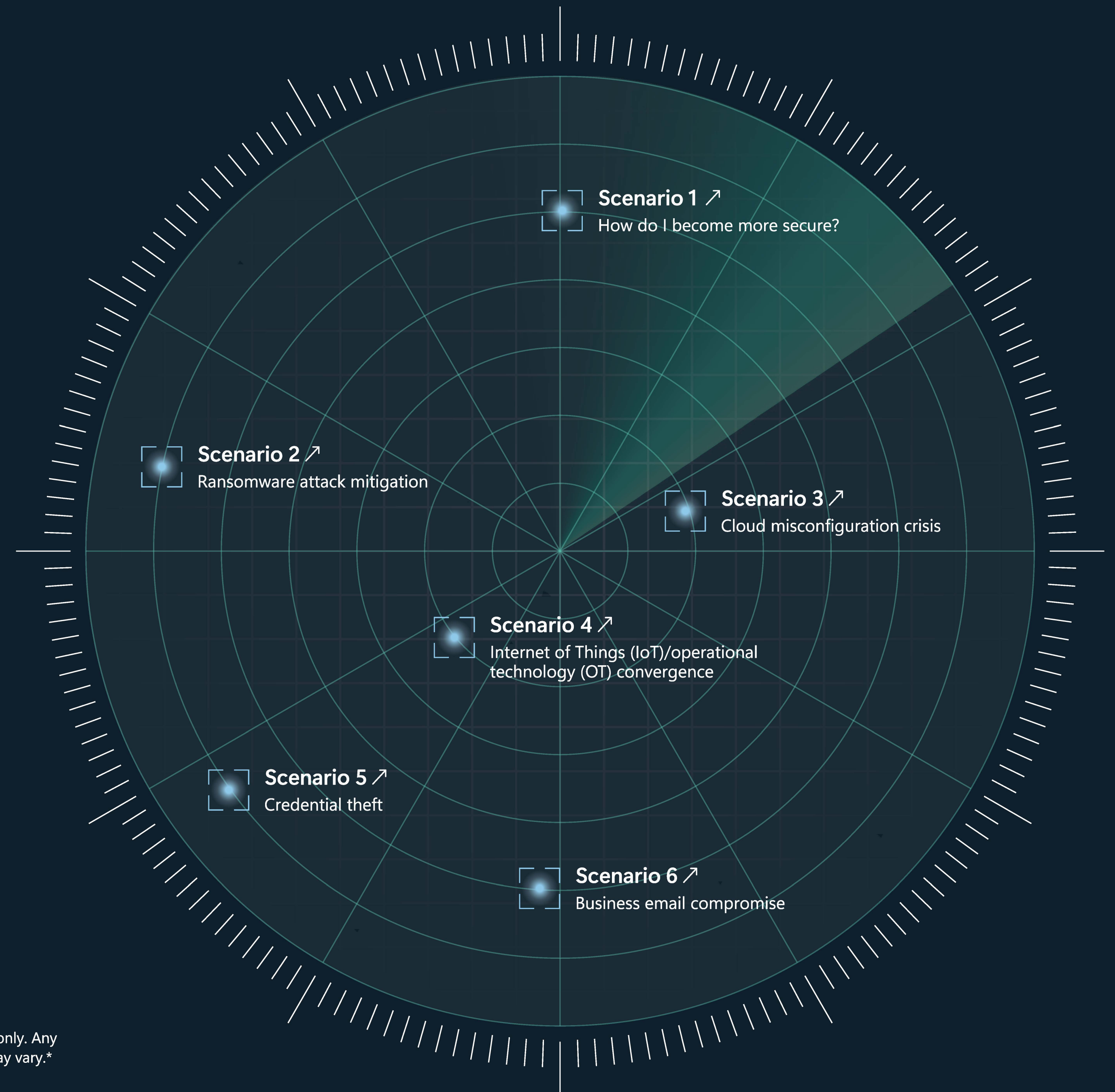
Navigating this interactive guide

Just as a ship's radar detects storms over open waters, Microsoft Security Exposure Management helps organizations spot cyber risks before they escalate. This interactive guide uses a **radar-inspired interface** to visualize your most critical security scenarios*—each representing a real-world threat pattern observed across Microsoft global threat intelligence.

How to proceed

- 1. Choose your storm**
Click any "risk zone" on the radar diagram below.
- 2. Understand the context**
Go through the risks that scenario brings to an organization.
- 3. Find out how Microsoft Security Management can help**
Get Microsoft Security Exposure Management responses.
- 4. Discover the outcomes**
Learn how our platform can aid your security team.
- 5. Course-correct anytime**
Return to radar view via the "Return home" button.

The companies and scenarios depicted in this content are entirely fictional and created for demonstration purposes only. Any resemblance to real companies or actual customer experiences is purely coincidental. Actual customer experiences may vary.



					
Scenario 1	Scenario 2	Scenario 3	Scenario 4	Scenario 5	Scenario 6
How do I become more secure?	Ransomware attack mitigation	Cloud misconfiguration crisis	IoT/OT convergence risks	Credential theft	Business email compromise

SCENARIO 1

How do I become more secure?

During a routine inspection, Global TechForward Solutions, a rapidly growing tech company, identified critical exposures during a security assessment. It found:

- Unencrypted sensitive data in the cloud.
- Outdated firewall rules on IoT devices.
- Excessive user access to critical systems.
- High employee susceptibility to simulated phishing attacks.

The CISO needed to remediate risks within 90 days to maintain compliance and avoid penalties.



Read on to discover what Microsoft Security Exposure Management can do

How do I become more secure?

What can Microsoft Security Exposure Management do?

Unified view of security posture

Security Exposure Management provides a comprehensive view of security posture across company assets and workloads, continuously discovering and gathering data into an up-to-date inventory of your attack surface.

Attack surface management

Visualize, analyze, and manage cross-workload attack surfaces using the enterprise exposure graph and attack surface map.

Critical asset protection

Identify and safeguard critical assets, enabling focused prioritization to ensure security and business continuity.

Exposure insights and management

Take advantage of aggregated data and rich context to prioritize security efforts. Use actionable recommendations to mitigate identified attack paths.

Continuous threat exposure management (CTEM)

Build an effective exposure management program with a CTEM process, reducing risk through real-time assessment of potential exposures

Outcomes

Secured critical assets with clear visibility

The Microsoft Security Exposure Management dashboard transformed the organization's reactive approach into proactive defense. The CISO tracked progress in real time, prioritizing fixes like encrypting cloud storage and hardening IoT firewall rules and key vulnerabilities from their initial audit.

Automated compliance alignment

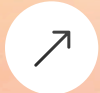
By gaining visibility into critical attack paths through the Microsoft Security Exposure Management dashboard, the team surgically removed choke points like overprivileged service accounts and exposed cloud storage buckets. This enabled manual prioritization of firewall rule hardening and encryption implementation.



Scenario 1
How do I **become** more secure?



Scenario 2
Ransomware attack mitigation



Scenario 3
Cloud misconfiguration crisis



Scenario 4
IoT/OT convergence risks



Scenario 5
Credential theft



Scenario 6
Business email compromise

SCENARIO 2

Ransomware attack mitigation

Global Healthcare Corp recently detected unusual file encryption activity in its research division, aligning with tactics used by the “Octo Tempest” ransomware group. The attackers began with a phishing email carrying hidden macros, targeting busy executives who frequently handle critical business documents. Once a single endpoint was compromised, lateral movement escalated, allowing the attackers to encrypt sensitive data and demand a hefty ransom in cryptocurrency. Without proactive exposure management, the corporation risked massive financial loss, operational disruption, and damage to client trust.



Read on to discover what
Microsoft Security Exposure
Management can do

Ransomware attack mitigation

What can Microsoft Security Exposure Management do?

Endpoint vulnerability analysis

Continuously scan endpoints to detect unauthorized scripts, unpatched applications, and hidden ransomware indicators. Then, integrate signals from Microsoft Defender for Endpoint and Microsoft 365 Defender to compile a prioritized remediation list.

Threat intelligence correlation

Take advantage of global threat intelligence feeds to identify malicious IP addresses, phishing domains, or known ransomware behaviors associated with Octo Tempest.

Attack path reduction

Microsoft Security Exposure Management identifies risks such as privilege escalation, external reachability, over permissions, policy misconfigurations, and visibility blind spots.

Mitigation guidance

Provide step-by-step recommendations to isolate infected systems, update security policies, and strengthen email hygiene configurations.

Outcomes

Contained encryption spread

Visibility from the Microsoft Security Exposure Management platform enabled Global Finance Corp to reduce ransomware exposure. This allowed the organization to see the risk of certain unpatched servers, prompting swift containment and halting adversarial lateral movement.

Protected critical care systems

Microsoft Security Exposure Management surfaced IoT/OT segmentation vulnerabilities, enabling policy adjustments that kept ICU monitoring devices isolated from ransomware infiltration. Continuous exposure monitoring ensured no downtime for life-critical systems, maintaining operational stability.

Cost-effective recovery

Microsoft Security Exposure Management reports flagged nine high-priority Common Vulnerabilities and Exposures, leading to immediate patching that saved an estimated USD2.1 million in potential recovery costs. This prioritized remediation workflow cut incident response times, reducing forensics overhead.



Scenario 1

How do I become more secure?



Scenario 2

Ransomware attack mitigation



Scenario 3

Cloud misconfiguration crisis



Scenario 4

IoT/OT convergence risks



Scenario 5

Credential theft



Scenario 6

Business email compromise

Return home



SCENARIO 3

Cloud misconfiguration crisis

During a routine audit, Global Energy Corp discovered that its customer data storage service (housed in a public cloud environment) had overly permissive access controls. Files holding account information could be queried without proper credentials. Cybercriminals exploited this gap and exfiltrated sensitive details before the breach was detected. The aftermath included legal scrutiny, compliance violations, and a spike in identity theft concerns among affected clients.



Read on to discover what Microsoft Security Exposure Management can do

Cloud misconfiguration crisis

What can Microsoft Security Exposure Management do?

Real-time monitoring of cloud configurations

Continuously evaluate Microsoft Azure and Microsoft 365 environments for dangerous or excessive permissions, open public endpoints, and missing encryption settings.

Remediation workflow automation

Generate prioritized tasks to rectify misconfigurations (for example, restricting network access, enforcing encryption at rest, and ensuring identity-based access).

Compliance and governance alignment

Map discovered misconfigurations against regulatory requirements, such as ISO 27001 or PCI DSS to ensure immediate corrective actions.

Continuous verification

Implement policy checks to make sure any changes to cloud services remain within approved security baselines.

Outcomes

Accelerated threat detection

Microsoft Security Exposure Management integrated with cloud security posture scans to reduce misconfiguration detection time by 79% (48 hours → 10 hours). Proactive alerts on public endpoints and weak authentication measures triggered immediate lockdown procedures that mitigated the risk of a repeat of the incident.

Intellectual property protection

Automated storage hardening, driven by Microsoft Security Exposure Managements prioritized remediation tasks, safeguarded USD150 million worth of AI training data. Microsoft Security Exposure Management analysis and subsequent Microsoft Security Experts–provided recommendations on access control and encryption curbed unauthorized access attempts, ensuring data integrity.

Cost optimization

Just-in-time access suggestions from Microsoft Security Exposure Management reduced overprivileged service accounts by 89%, lowering cloud resource misuse. This helped optimize identity management, and role assignments diminished cross-tenant complexities.

					
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SCENARIO 4

IoT/OT convergence risks

To modernize, Global Mining Corp integrated IoT devices into its corporate office, including smart cameras, HVAC systems, and connected conference room technologies. Some business units merged these IoT systems with OT, controlling core functions like power management. A single IoT device with weak firmware became a pivot point, allowing attackers to jump from facility controls into sensitive corporate networks.



Read on to discover what Microsoft Security Exposure Management can do

IoT/OT convergence risks

What can Microsoft Security Exposure Management do?

Discovery and classification

Continuously scan for newly added IoT and OT devices, ensuring each is correctly classified and tracked.

Firmware and configuration analysis

Spot unpatched firmware, default passwords, or outdated protocols that make IoT systems an easy target.

Automated threat modeling

Map potential attack paths from compromised IoT devices to financial back-end systems and highlight priority remediation steps.

Outcomes

Critical infrastructure protection

Microsoft Security Exposure management helped to map IoT/OT assets and flagged priority vulnerabilities, ensuring FIPS 140-2 encryption across 450 field devices to prevent grid manipulation attacks. This consistent and continuous scanning identified out-of-date firmware, prompting timely patching to stop potential exploits.

Unified visibility

Microsoft Defender for IoT insights consolidated attack pathway mapping, discovering 122 unmanaged OT assets critical to hybrid attack chains. This single-pane-of-glass approach enabled security teams to disrupt lateral movement vectors across industrial and enterprise endpoints through exposure management protocols.

Cost avoidance

Proactive risk modeling prevented minor vulnerabilities from snowballing into major incidents.

Regulatory alignment

Ongoing scans kept the organization's compliance posture up to date, easing regulatory audits. This is because Microsoft Security Exposure Management allowed defenders to automap identified risks to NERC CIP v7 standards, guiding swift remediation for ICS access control gaps.

					
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SCENARIO 5

Credential theft

A series of password spray and brute-force attacks targeted employee accounts at Global Retail Corp. Once attackers cracked even a single set of valid credentials, they exploited them to move laterally, escalate privileges, and run silent data exfiltration campaigns. Lacking continuous monitoring, the organization remained unaware of the unauthorized logins until unusual transaction logs sparked an audit.



Read on to discover what Microsoft Security Exposure Management can do

Credential theft

What can Microsoft Security Exposure Management do?

Identity exposure discovery

Scan Active Directory, Azure AD, and Microsoft Entra ID to identify insecure password policies, weak or non-existent multifactor authentication (MFA) configurations, and dormant accounts susceptible to brute-force attacks.

Adaptive access policies

Recommend conditional access rules that dynamically adjust authentication requirements based on risk signals, such as location or device compliance.

Streamlined forensics

Correlate sign-in patterns across platforms to spot anomalies faster, offering quick pivot-to-investigation workflows.

Outcomes

Fraud prevention

Microsoft Security Exposure Management helped simulation tools identify key phishing risk behaviors.

Compliance achievement

End-to-end visibility proved pivotal in meeting critical data protection mandates across retail operations.

Financial protection

Secure rotation of compromised service accounts curtailed lateral movement and data exfiltration attempts.

					
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SCENARIO 6

Business email compromise

Global Finance Corp managed billions in transactions daily across various clients and partners. With an expanding remote workforce and multiple business units relying heavily on email communication, the organization became vulnerable to business email compromise (BEC) attacks.

An attacker successfully impersonated a high-level executive by exploiting an exposed email configuration misalignment and weak MFA enforcement. The fraudulent email instructed the finance department to process

an urgent wire transfer to an external account. Due to lack of visibility into identity vulnerabilities and misconfigurations, the finance team unknowingly complied, resulting in a substantial financial loss.

Without proactive exposure management, Global Finance Corp risked ongoing BEC threats, potential regulatory fines, reputational damage, and erosion of customer trust.



Read on to discover what Microsoft Security Exposure Management can do

Business email compromise (BEC)

What can Microsoft Security Exposure Management do?

Comprehensive identity exposure assessment

Microsoft Security Exposure Management continuously scans for misconfigurations in identity systems, such as weak MFA policies, excessive permissions, and risky forwarding rules that could be exploited in BEC attacks. Pulling from both Microsoft Defender for Office 365 and Microsoft Entra workloads.

Real-time risk prioritization

By analyzing both internal identity vulnerabilities and external threat intelligence, it identifies which accounts or systems are most susceptible to BEC-related social engineering tactics.

Attack path simulation

Maps potential attack paths that could lead from compromised email accounts to sensitive financial systems, highlighting critical choke points where attackers could escalate privileges or manipulate workflows.

Outcomes

Reduced risk of financial fraud

Early detection of identity exposures and risky configurations prevented unauthorized access to critical business email accounts.

Improved security posture

Continuous scoring and visibility into identity-related risks enable security teams to prioritize high-impact actions, ensuring critical vulnerabilities were addressed promptly.

Enhanced incident response

Clear mapping of potential attack paths streamlined incident response efforts, reducing the time and resources needed to investigate and remediate BEC attempts.

Sustained trust and compliance

Regular reporting showcased ongoing security improvements, supporting regulatory compliance and reinforcing stakeholder confidence in the organization's ability to safeguard sensitive information and assets.

Conclusion and resources

Charting a course through today's cybersecurity landscape demands vigilance, adaptability, and the right set of integrated tools. Like a seasoned ship's captain scanning the horizon, Microsoft Security Exposure Management empowers organizations to spot looming threats early and navigate confidently. Its comprehensive, cross-domain insights serve as a guiding compass, effectively steering around digital storms.

Ready to implement Microsoft Security Exposure Management? Visit [Microsoft Security Exposure Management](#) or contact your account manager for more details today.

Resources

- ➞ **Microsoft Security Blog**
Authoritative insights, best practices, and updates on Microsoft Security's approach to cybersecurity.
- ➞ **Microsoft Security Exposure Management Overview**
Official outline of exposure management capabilities and risk assessment strategies.

